Solution Requirements

Date	29 June 2026
Team ID	SWTID-2026-4836
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** Authentication – Farmers can securely log in.
- **Non-Functional Requirements (NFR)** Performance – Prediction in less than 2 seconds.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	*Farmers can securely log in using username and password to access crop recommendations.*	High
2	Authorization levels	*Admin can manage datasets and models, while farmers can only view crop recommendations.*	Medium

3	External interfaces	*The system uses agricultural datasets stored in CSV files and machine learning libraries such as Scikit-learn and Pandas.*	High
4	Transactions processing	*The system receives soil and environmental parameters, processes them through the ML model, and returns suitable crop recommendations.*	High
5	Reporting	*The system displays predicted crops, analysis reports, and data visualizations to users.*	Medium
6	Business rules, etc.	*Crop recommendations are generated only when all required parameters are provided and validated.*	High
7	Compliance to laws or regulations	*Crop recommendations are generated only when all required parameters are provided and validated.*	Low
8	*Other*	*The system allows future integration with weather APIs and real-time agricultural data sources.*	Low

Step 2: Non-Functional Requirements (NFR)

S.NO	REQUIREMENT CATEGORY	Requirement Description	Priority (High/Medium/Low)
1	Performance & Speed	*The system should generate crop recommendations quickly*	*Prediction generated within 2 seconds.*
2	Scalability	*The system should support an increasing amount of agricultural data and users.*	*Supports more than 1000 users and large datasets.*
3	Security & Data Privacy	*User data and prediction records should be protected from unauthorized access.*	*Password protection and secure storage of records.)*
4	Reliability & Availability	*The application should work continuously without frequent failures.*	*99% system availability.*
5	Usability & Accessibility	*The interface should be simple and easy for farmers to use.*	*User-friendly interface with clear instructions.*
6	*Other*	*The system should be easy to update and run on different operating systems.*	Supports Windows, Linux, and macOS.